disease, but stimulate the body to produce antibodies against the antigen in the microbes.

Table 22.2 Vaccines produced against diseases

Type of Vaccine	Name of the vaccine	Disease
Live attenuated	MMR	Measles, Mumps and Rubella
	BCG (Bacillus Calmette Guerin)	Tuberculosis
Inactivated (Killed antigen)	Inactivated polio virus (IPV)	Polio
Subunit vaccines (Purified antigens)	Hepatitis B vaccine	Hepatitis B
Toxoid (Inactivated antigen)	Tetanus toxoid (TT)	Tetanus
	Diphtheria toxoid	Diphtheria

22.3 Microbes and Diseases

Disease (dis = against; ease = comfort) can be defined as an impairment or malfunctioning of the normal state of the living organism that disturbs or modifies the performance of vital functions of the body. Disease can be categorized based on:

i. The extent of occurrence (endemic, epidemic, pandemic or sporadic).
ii. Whether infectious or non-infectious.
iii. Types of pathogen – whether caused by bacterial, viral, fungal or protozoan infections.
iv. Transmitting agent – whether air borne, water borne or vector borne.

DO YOU KNOW?

World Health Day – 7th April
World Malaria Day – 25th April
World AIDS Day – 1st December
World Anti -Tuberculosis Day – 24th March

22.3.1 Classification of Disease based on Occurence

Endemic: Disease which is found in a certain geographical area affecting a fewer number of people (low incidence). e.g. Occurrence of goitre in Sub-Himalayan regions.

Epidemic: Disease which breaks out and affects large number of people in a particular geographical region and spreads at the same time. e.g. Influenza.

Pandemic: Disease which is widely distributed on a global scale. e.g. Acquired Immuno deficiency Syndrome (AIDS).

Sporadic: Disease which occur occasionally. e.g. Malaria and Cholera.

22.3.2 Manifestation of Disease

Communicability of diseases

Infectious diseases are communicable diseases. They are caused by external factors like pathogenic organisms (bacteria, virus, vectors, parasites) invading the body and causing diseases. e.g. Influenza, Tuberculosis, Chickenpox, Cholera, Pneumonia, Malaria, etc

Non-infectious diseases are non-communicable diseases. They are caused by internal factors like malfunctioning of organs, genetic causes, hormonal imbalance and immune system defect. e.g. Diabetes, Coronary heart diseases, Obesity, Cancer, Goitre, etc

Point of entry and place of infection

The disease causing microbes enter the body through different means. An infection develops when these pathogens enter the human body through contaminated air, water, food, soil, physical contact, sexual contact and through infected animals. They may be organ specific or tissue specific within our body where microbes reside.

Reservoir of infection

Reservoir of infection refers to the specific environment in which the pathogens can thrive well and multiply without causing diseases. eg. Water, soil and animal population.

Incubation period

The interval between infection and first appearance of the diseases is called incubation period. It may vary from few hours to several days.

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